

# Deferred-Brick Mirror-Fork Analysis & Safe-Convergence Brief

**Revision:** 1 **Last modified:** 2026-07-09T20:33:57Z **Author:** Stream MF  
(autonomous READ-ONLY investigation subagent) **Scope:** submodules/  
llm\_orchestrator + submodules/vision\_engine **Constraint authority:**  
§11.4.113 (absolute no-force-push, merge-onto-latest-main) · §9.2 (zero  
commit loss) · §11.4.6 (no guessing)

---

## 0. Executive summary

Two owned Go bricks were deferred by the gofmt sweep because their  
git **mirrors have diverged** — a pre-existing fork, NOT caused by the  
sweep. This brief establishes the exact fork facts (bidirectional, with  
authorship + content), determines the safe lossless convergence per  
§11.4.113, and states the single operator decision each needs.

| Brick            | origin<br>unique                  | github/<br>upstream<br>ahead by | Merge<br>onto<br>github<br>base                                                    | Commit<br>loss? | gofmt<br>residual<br>(github<br>tip) |
|------------------|-----------------------------------|---------------------------------|------------------------------------------------------------------------------------|-----------------|--------------------------------------|
| llm_orchestrator | 1<br>commit<br>(a484f7d,<br>ours) | +34                             | add/add<br>CONFLICT<br>on helix-<br>deps.yaml<br>(trivial,<br>content<br>decision) | None            | 0 files —<br>sweep is<br>a no-op     |
| vision_engine    | 1<br>commit<br>(b417a40,<br>ours) | +12                             | CLEAN<br>(disjoint<br>paths, no<br>conflict)                                       | None            | 17 files<br>— sweep<br>applies       |

| Brick | origin<br>unique | github/<br>upstream<br>ahead by | Merge<br>onto<br>github<br>base | Commit<br>loss? | gofmt<br>residual<br>(github<br>tip)<br><br>after<br>converge |
|-------|------------------|---------------------------------|---------------------------------|-----------------|---------------------------------------------------------------|
|-------|------------------|---------------------------------|---------------------------------|-----------------|---------------------------------------------------------------|

Both origin-unique commits are **ours** (author `i@mvasic.ru`, Milos Vasic / Милош Васић), **real changes** (not merge artifacts), and are **preserved** by the recommended merge — nothing is lost in either direction.  
**Recommended canonical base for both: the github/upstream tip**  
(the most-advanced mirror; identical `github==upstream`).

Neither convergence requires — nor permits — a force-push. Both are fast-forward pushes after the merge because the merge commit descends from every mirror tip.

# 1. Shared facts (verified 2026-07-09, read-only `git fetch --all --prune`)

Both bricks are on detached HEAD at their parent-pinned commit.  
Remotes for each:

- `origin fetch` = `vasic-digital/<Repo> · origin push` = `HelixDevelopment/<Repo>`
- `github` = `upstream` = `HelixDevelopment/<Repo>` (identical tips)

| Brick            | origin/<br>master | github/<br>master =<br>upstream/<br>master | Parent<br>pins | Parent<br>pin is...                    |
|------------------|-------------------|--------------------------------------------|----------------|----------------------------------------|
| llm_orchestrator | a484f7d           | ee229a7                                    | a484f7d        | the <b>origin tip</b>                  |
| vision_engine    | b417a40           | a97df79                                    | 0bf75ee        | the <b>merge-base</b><br>(behind BOTH) |

## 2. llm\_orchestrator (branch master)

### 2.1 Fork geometry

- **Common ancestor (merge-base):** d2a2151 — “*cascade: chore(1.1.8-dev-rc1 cascade)*...”, Милош Васић, 2026-06-02.
- `git merge-base --is-ancestor origin github` → **NO** (genuine divergence, not a stale-behind).
- **origin = merge-base + 1 commit; github = merge-base + 34 commits.**

### 2.2 origin's unique commit github lacks (github..origin) — exactly ONE

```
a484f7d chore(deps): add §11.4.31 helix-deps.yaml dependency
manifest
    Author: Milos Vasic <i@mvasic.ru>    Date: 2026-06-08
21:23:28 +0500
    Parent: d2a2151 (the merge-base)
    Adds: helix-deps.yaml | 32 insertions (new file)    Co-
Authored-By: Claude Opus 4.8
```

**Ours / foreign:** OURS (i@mvasic.ru). **Real change / artifact:** real change — a new 32-line helix-deps.yaml declaring the LLMProvider own-org dep (git@github.com:vasic-digital/LLMProvider.git, ref: main, flat layout).

### 2.3 What github has that origin lacks (origin..github) — 34 commits

github is a genuine, far-more-advanced line: a CONSTITUTION.md full rewrite + wrong-project (“HelixCode”) decontamination, §11.4.29 lowercase-upstreams migration, anti-bluff test hardening, §11.4.103-165 governance cascade, real bug-fixes (a446a3d protocol ctx-cancel message loss, 4e08b93 RoundRobinSelector, 1fac8b1 SimpleAgentPool.Release, c6d6049 opencode Wait() hang), **and 974d784 “style: gofmt -w hygiene across llm\_orchestrator”** — i.e. the gofmt work the sweep intended is **already on github’s line**. All authored by our org. This is **not** a foreign line — it is our own, later, decontaminated + gofmt-clean trunk that origin never received.

## 2.4 The conflict (verified via `git merge-tree --write-tree github origin`)

CONFLICT (add/add): Merge conflict in helix-deps.yaml

### Both mirrors independently added helix-deps.yaml: - origin

(a484f7d, 2026-06-08, 32 lines): declares LLMProvider as an own-org dep.

- **github** (3ac8124, 2026-06-17, 8 lines): deps: [] with a §11.4.74

catalogue-check note “*own-org deps: none*”.

These two files **semantically contradict** (LLMProvider-is-a-dep vs no-own-org-dep). This is the one point that needs a human decision — see §4.1. It is a **content reconciliation**, NOT a commit loss: merging preserves both a484f7d and all 34 github commits in history; only the file’s final bytes are chosen.

Honest boundary (§11.4.6): I did **not** open `go.mod` to decide which manifest is factually correct (whether LLMProvider is still a live replace). That is the operator/author call flagged in §4.1 — I state it as UNKNOWN rather than guess.

## 2.5 gofmt residual

github tip ee229a7 → gofmt -l = **0 files**. Once converged onto the github base the gofmt sweep is a **no-op** for this brick.

---

# 3. vision\_engine (branch master)

## 3.1 Fork geometry

- **Common ancestor (merge-base) = the parent pin 0bf75ee** — “Merge remote-tracking branch ‘origin/master’”, Милош Васић, 2026-06-16. Ancestor of BOTH mirrors (verified).
- `git merge-base --is-ancestor origin github` → **NO** (genuine divergence).
- **origin = merge-base + 1; github = merge-base + 12**. The parent **currently pins the common ancestor** — behind github by 12 and behind origin by 1.

### 3.2 origin's unique commit github lacks (github..origin) — exactly ONE

```
b417a40 chore(governance): add install_upstreams.sh + Upstreams/
{GitHub,GitLab} (§11.4.36 / §6.W)
```

```
Author: Милош Васић <i@mvasic.ru> Date: 2026-07-04
```

```
16:45:18 +0300
```

```
Parent: 0bf75ee (the merge-base)
```

```
Adds: Upstreams/GitHub.sh (9) + Upstreams/GitLab.sh (9) +
install_upstreams.sh (109) = 127 insertions
```

**Ours / foreign:** OURS (i@mvasic.ru). **Real change / artifact:** real change — adds a root `install_upstreams.sh` (not present at github tip) plus **capital-Upstreams/** recipe scripts.

### 3.3 What github has that origin lacks (origin..github) — 12 commits

Our own later line: wrong-project decontamination, §11.4.29

**lowercase upstreams/ migration** (9553a31 “lowercase upstream scripts; simplify push-all.sh”), anti-bluff test hardening, Dockerfile non-root hardening (f3ef1f2), §11.4.142-165 cascade. All i@mvasic.ru / our org. Genuine trunk, not foreign.

### 3.4 The merge (verified via `git merge-tree --write-tree github origin`)

```
229c431e59f16d5bced0b2fb64beb6ef82504a60 ← tree OID only, NO
conflict lines, exit 0
```

**CLEAN merge — zero conflicts.** origin's b417a40 touches `Upstreams/*` (capital U) + `install_upstreams.sh`; github's line touches `upstreams/*` (lowercase). On the case-sensitive Linux host these are **disjoint paths**, so git auto-merges cleanly and both lines' commits are preserved.

**Caveat for the operator (not a blocker):** the clean merge reintroduces the **capital-Upstreams/ directory**, which is a §11.4.29 **lowercase-naming concern** — the github line deliberately migrated to lowercase `upstreams/`. `install_upstreams.sh` at root is genuinely new and wanted (§11.4.36); the capital `Upstreams/GitHub.sh` + `GitLab.sh` are redundant with the existing lowercase recipes. See §4.2.

### 3.5 gofmt residual

github tip a97df79 → gofmt -l = **17 files** (identical set on origin tip — the gofmt hygiene was never done on this brick's line): cmd/  
visiondescribe/main.go, pkg/analyzer/  
{i18n\_defaults,types,types\_test}.go, pkg/config/  
i18n\_callsites\_test.go, pkg/graph/{graph,graph\_automation\_test}.go,  
pkg/llmvision/  
{anthropic,astica,i18n\_defaults,provider,provider\_test}.go, pkg/  
opencv/{i18n\_defaults,orb\_vision\_test}.go, pkg/remote/  
{deployer\_test,remote,remote\_test}.go. The **gofmt sweep still applies to this brick and must run AFTER convergence** (on the merged tree).

---

## 4. Operator decision needed

### 4.1 llm\_orchestrator — resolve the helix-deps.yaml add/add conflict

The only real decision. Two contradictory manifests exist: - **origin a484f7d**: LLMProvider IS an own-org dep (via go.mod replace). - **github 3ac8124**: deps: [] — own-org deps: none.

Operator (or brick author) must confirm — from the brick's current go.mod — which is factually correct, then the merge keeps that content. If LLMProvider is still a live replace, keep origin's richer 32-line manifest; if the dep was removed, keep github's deps: []. **UNKNOWN to this read-only investigation — flagged, not guessed (§11.4.6).**

### 4.2 vision\_engine — capital Upstreams/ vs lowercase upstreams/

The merge is clean but reintroduces capital Upstreams/. Operator chooses one: - **(a)** Keep install\_upstreams.sh (wanted, §11.4.36) but **drop / lowercase-rename** the capital Upstreams/GitHub.sh+GitLab.sh during the merge to honour §11.4.29 (the github line already has lowercase equivalents) — **recommended**; or - **(b)** Accept the capital dir as-is (clean auto-merge, but a standing §11.4.29 nit).

---

## 5. Recommended safe convergence (§11.4.113 — merge-onto-latest, FF-push, NEVER force)

**Canonical base for BOTH bricks: the github/upstream tip** (most-advanced mirror; github==upstream; already decontaminated; llm\_orchestrator already gofmt-clean).

Per-brick procedure (executed by the conductor, not this read-only stream):

1. `git fetch --all --prune --tags` (all remotes).
2. Check out / base on the **github tip** (ee229a7 / a97df79).
3. **Merge the origin unique commit** on top:
  - `llm_orchestrator`: merge a484f7d; resolve the `helix-deps.yaml` add/add per §4.1 (no -s ours, no drop — union history, chosen content).
  - `vision_engine`: merge b417a40 (auto-clean); apply the §4.2 choice.
4. `vision_engine` only: **run the gofmt sweep** on the merged tree (17 files) and commit.
5. Audit: no conflict markers, no file silently dropped, prior tests still green.
6. **Fast-forward push to ALL upstreams** (origin push → HelixDevelopment, github, upstream, + GitLab/GitFlic/GitVerse fan-out). Each push is a fast-forward because the merge commit descends from every mirror tip — **no force is ever needed**.
7. Advance the parent `.gitmodules` pointer to the new merged tip in the same change window.

**Commit-loss check (§9.2):** NONE in either direction. The merge preserves origin's 1 commit AND github's 34 / 12 commits. The recommended base flips each parent pin FORWARD (`llm_orchestrator a484f7d` → merged, `vision_engine 0bf75ee` → merged) with zero history rewrite.

**Force-push:** not required and forbidden (§11.4.113) — the merge-onto-latest path makes every push a fast-forward.

---

## 6. Verification trail (read-only commands run)

- `git fetch --all --prune` (both bricks) — remote tips captured.
- `git merge-base / --is-ancestor` — divergence + ancestor facts.
- `git log --oneline <a>..<b>` both directions — unique-commit sets.
- `git show --stat --format=...` — authorship + content of each origin-unique commit.
- `git ls-tree -r --name-only <tip>` — file presence (helix-deps.yaml, upstreams paths).
- `git merge-tree --write-tree <github> <origin>` — conflict simulation (no working-tree or ref mutation): `llm_orchestrator = add/add CONFLICT; vision_engine = clean`.
- `git archive <tip> | tar -x to scratchpad + gofmt -l` — residual gofmt (0 / 17), scratch cleaned after.

No ref, file, or working tree in either submodule or the parent was modified. Only this document was written.